



Grade 4
General
Mathematics



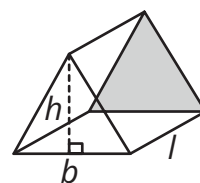
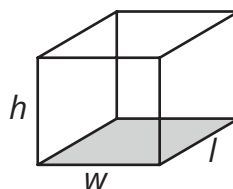
NSCAS NEBRASKA STUDENT-CENTERED ASSESSMENT SYSTEM

Mathematics Reference Sheet

2-Dimensional Shape	Area	Circumference
Circle	$A = \pi r^2$	$C = \pi d = 2\pi r$
Triangle	$A = \frac{1}{2}bh$	Perimeter
Rectangle	$A = l \times w$	$P = 2l + 2w = 2(l + w)$
Square	$A = s \times s$	$P = s + s + s + s$
Trapezoid	$A = \frac{1}{2}h(b_1 + b_2)$	
Parallelogram	$A = bh$	

3-Dimensional Shape	Volume	Surface Area
Rectangular Prism	$V = lwh = Bh$	SA = the sum of the areas of all the faces
Triangular Prism	$V = \frac{1}{2}bhl = Bl$	SA = the sum of the areas of all the faces
Cone	$V = \frac{1}{3}\pi r^2h$	
Cylinder	$V = \pi r^2h$	
Sphere	$V = \frac{4}{3}\pi r^3$	

In the images to the right, the shaded faces are the bases.



2-Dimensional Shape Key		
b = base length	l = length	s = side
h = height	w = width	
d = diameter	r = radius	

Use 3.14 for π .

3-Dimensional Shape Key		
d = diameter	l = length	w = width
r = radius	h = height of shape	B = area of base
	b = base length	

Percent Change	Pythagorean Theorem
% change = $\frac{\text{difference in amounts}}{\text{original amount}}$	$c^2 = a^2 + b^2$

Standard Units	Metric Units
Conversions – Length	
1 foot (ft) = 12 inches (in.)	1 centimeter (cm) = 10 millimeters (mm)
1 yard (yd) = 3 feet (ft) = 36 inches (in.)	1 meter (m) = 100 centimeters (cm)
1 mile (mi) = 1,760 yards (yd) = 5,280 feet (ft)	1 meter (m) = 1,000 millimeters (mm)
	1 kilometer (km) = 1,000 meters (m)
Conversions – Volume	
1 cup = 8 fluid ounces (fl oz)	1 liter (l) = 1,000 milliliters (ml)
1 pint (pt) = 2 cups	1 liter (l) = 1,000 cubic centimeters (cu cm)
1 quart (qt) = 2 pints (pt)	
1 gallon (gal) = 4 quarts (qt)	
Conversions – Weight/Mass	
1 pound (lb) = 16 ounces (oz)	1 gram (g) = 1,000 milligrams (mg)
1 ton = 2,000 pounds (lb)	1 kilogram (kg) = 1,000 grams (g)

Directions:

On the following pages of your practice test booklet are questions for the Grade 4 *Nebraska Student-Centered Assessment System (NSCAS) General Mathematics* practice test.

Read these directions carefully before beginning the test.

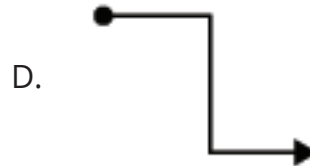
This practice test will include several different types of questions. Multiple choice questions will ask you to select an answer from among four choices. Multiple select questions will ask you to select multiple correct answers from among five or more answer choices. For some questions, there may be two parts, Part A and Part B, where each part has a multiple choice or multiple select question. Some questions will ask you to construct an answer by following the directions given.

For all questions:

- Read each question carefully and choose the best answer.
- You may use scratch paper to solve the problems.
- The Mathematics Reference Sheet is provided in the back of the test booklet. You may refer to this page at any time during the test.
- Be sure to answer ALL the questions.

1. What is the standard form of forty-five and nine tenths?
 - A. 45.009
 - B. 45.09
 - C. 45.9
 - D. 45.910

2. Identify the figures that BEST show a line, a line segment, and a ray. Write the letters of the correct figure in the table.



Name	Line	Line Segment	Ray
Figure			

3. What is the product of $7 \times 1,658$?

- A. 7,256
- B. 7,606
- C. 11,565
- D. 11,606

4. Which set is ordered from least to greatest?

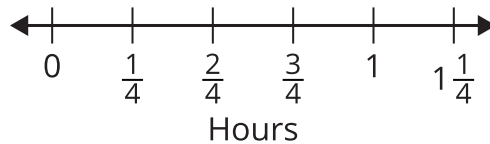
- A. $\frac{8}{9}, \frac{1}{2}, \frac{3}{4}, \frac{1}{10}$
- B. $\frac{1}{10}, \frac{3}{4}, \frac{1}{2}, \frac{8}{9}$
- C. $\frac{1}{10}, \frac{1}{2}, \frac{3}{4}, \frac{8}{9}$
- D. $\frac{8}{9}, \frac{1}{10}, \frac{3}{4}, \frac{1}{2}$

5. Use the table to answer the question.

Day of the Week	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Hours Spent Reading	1	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	1	$\frac{3}{4}$	$\frac{3}{4}$

Write **Xs** on the number line to create a line plot of the data.

Hours Spent Reading



6. Use the protractor tool to measure the angle below.



What is the measure of the angle?

- A. 35°
- B. 40°
- C. 45°
- D. 50°

7. This question has two parts. Answer Part A, and then answer Part B.

Part A

What is the decimal form of $\frac{8}{10}$?

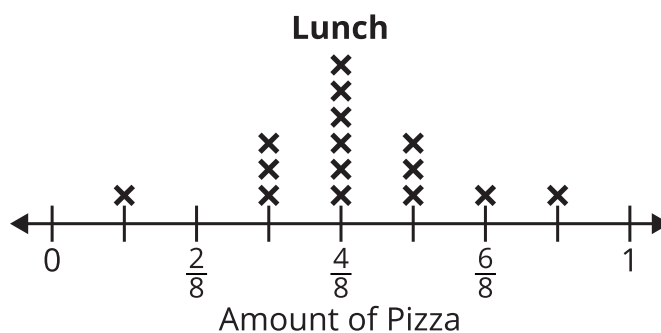
- A. 0.8
- B. 0.08
- C. 8.0
- D. 8.10

Part B

What is the decimal form of $\frac{38}{100}$? Write the answer in the box.

8. Oscar has 9 pumpkins. Libby has p fewer pumpkins than Oscar. Which expression represents the number of pumpkins Libby has?
- A. $9 + p$
 - B. $9 - p$
 - C. $p + 9$
 - D. $p - 9$

9. Use the line plot to complete the statements.



Students in a class ate pizza for lunch. The line plot shows the fraction of a pizza each student ate. Each **x** represents one student.

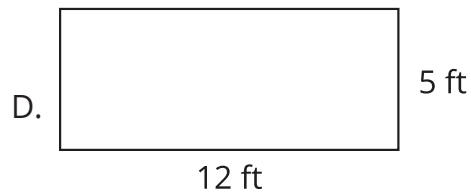
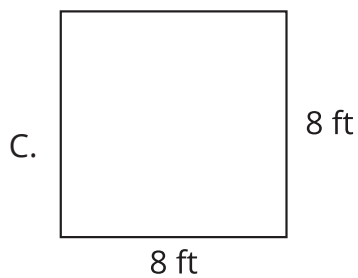
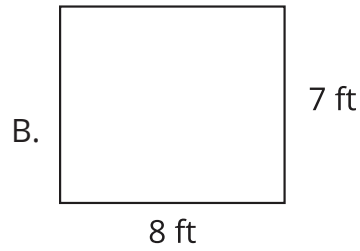
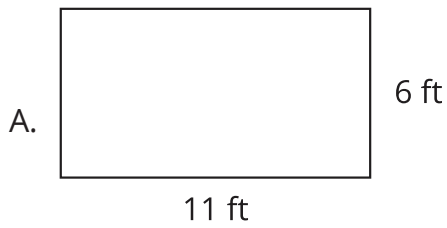
Select the correct amount of pizza to complete each sentence.

The two students who ate the most pizza ate a total of ($1\frac{4}{8}$ $1\frac{5}{8}$ $1\frac{6}{8}$ 2) pizzas.

The student who ate the most pizza ate ($\frac{4}{8}$ $\frac{6}{8}$ 1 6) pizza(s) more than the student who ate the least pizza.

The teacher ate $\frac{3}{8}$ of a pizza more than the student who ate the most pizza. The teacher ate ($1\frac{1}{8}$ $1\frac{2}{8}$ $1\frac{3}{8}$) pizza(s).

10. Which figure has the greatest area?



11. What is $4\frac{3}{10} + 8\frac{5}{10}$?

A. $12\frac{8}{100}$

B. $12\frac{15}{100}$

C. $12\frac{8}{20}$

D. $12\frac{8}{10}$

NSCAS GENERAL — MATHEMATICS

12. Which expressions have a remainder? Select the correct choice for each expression.

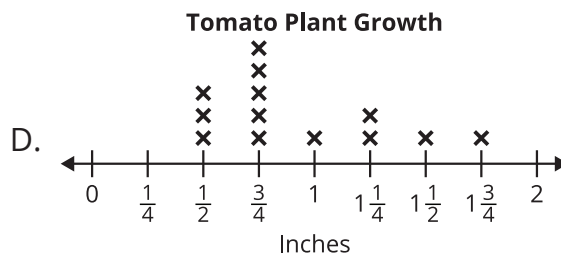
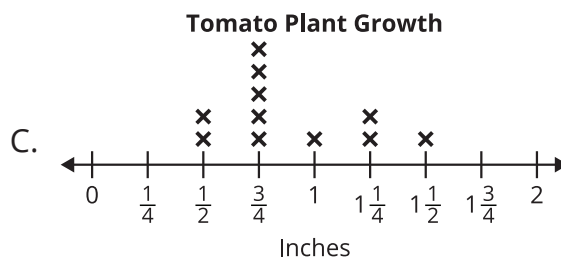
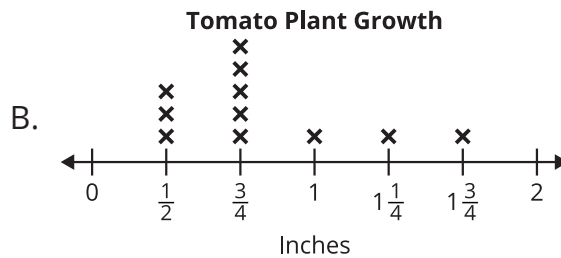
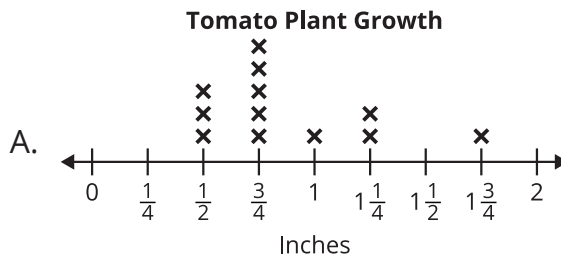
$948 \div 2$	Has a Remainder	Does Not Have a Remainder
$1,273 \div 3$	Has a Remainder	Does Not Have a Remainder
$2,700 \div 5$	Has a Remainder	Does Not Have a Remainder

13. A class recorded the growth, in inches, of some tomato plants over the past two days.

Tomato Plant Growth

$\frac{3}{4}$, $\frac{1}{2}$, $1\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$, $1\frac{3}{4}$, $\frac{3}{4}$, $\frac{3}{4}$, $1\frac{1}{4}$, $\frac{3}{4}$, 1

Which line plot shows the data from the tomato plants?



14. Which comparison is true?

- A. $4.09 > 4.50$
- B. $2.31 > 2.18$
- C. $5.23 < 5.14$
- D. $6.80 < 6.29$

15. **This question has two parts. Answer Part A, and then answer Part B.**

A school is ordering shelves for a teacher. The teacher has 17 math books, 18 science books, and 19 reading books. Each shelf holds 8 books.

Part A

How many shelves does the teacher need?

- A. 5 shelves
- B. 6 shelves
- C. 7 shelves
- D. 8 shelves

Part B

The teacher puts 8 books on each shelf except for the last shelf. How many books are on the last shelf?

- A. 7 books
- B. 6 books
- C. 4 books
- D. 0 books

16. How many inches are in 3 yards? Write the answer in the box.

inches

17. The table shows how far Mark rode his bike each day for three days.

Bike Riding

Day	Distance (miles)
Monday	$\frac{7}{10}$
Tuesday	$\frac{2}{10}$
Wednesday	$\frac{4}{10}$

What is the total distance Mark rode his bike?

- A. $\frac{5}{10}$ of a mile
- B. $\frac{9}{10}$ of a mile
- C. $1\frac{3}{10}$ miles
- D. $1\frac{13}{10}$ miles

18. Which shapes ALWAYS have two pairs of parallel sides?

Select **three** shapes.

- A. square
- B. rhombus
- C. trapezoid
- D. rectangle
- E. quadrilateral

19. What is the product of 18×24 ?

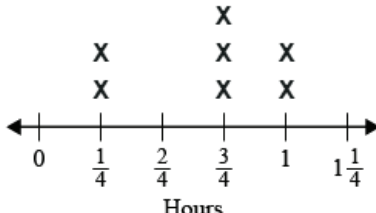
- A. 108
- B. 128
- C. 432
- D. 632

20. Which number is a multiple of 4?

- A. 30
- B. 38
- C. 42
- D. 48

**NSCAS General Grade 4
Practice Test Answer Key
Mathematics (English)**



Sequence	Key	Points								
1	C	1								
2	<table><tr><th>Name</th><th>Line</th><th>Line Segment</th><th>Ray</th></tr><tr><td>Figure</td><td></td><td></td><td></td></tr></table>	Name	Line	Line Segment	Ray	Figure				1
Name	Line	Line Segment	Ray							
Figure										
3	D	1								
4	C	1								
5	 Hours	1								
6	B	1								
7	Part A: A Part B: 0.38 or equivalent	2								
	Part A or Part B	1								
8	B	1								
9	1 5/8, 6/8, 1 2/8	2								
	2 of 3 correct responses	1								
10	A	1								
11	D	1								
12	Does Not Have a Remainder Has a Remainder Does Not Have a Remainder	1								
13	A	1								
14	B	1								
15	Part A: C Part B: B	2								
	Part A or Part B	1								
16	108 or equivalent	1								
17	C	1								
18	A, B, D	2								
	2 of 3 correct responses	1								
19	C	1								
20	D	1								